

AI COULD TRANSFORM OUR UNDERSTANDING OF ECOLOGICAL SYSTEMS

HARNESSING ARTIFICIAL INTELLIGENCE

IN CONTEXT

KEY ORGANIZATION

Cornell Lab of Ornithology (1915–)

BEFORE

1990s Scientists begin to use artificial intelligence (AI) algorithms as tools to classify land cover.

2002 Ornithologists at Cornell University's Lab of Ornithology, New York, launch eBird.

2010s AI revolutionizes the speed at which camera-trap data can be analyzed.

AFTER

2022 The UK Centre for Ecology and Hydrology develops a solar-powered autonomous station to record moths, birds, bats, and other mammals simultaneously.

2024 In Australia, AI-guided robots begin to implant larval corals into artificial "seed plugs" in order to accelerate reef restoration.

Tackling the biodiversity and climate crises requires data on organisms' population trends. Without it, ecologists cannot propose effective